

3	(a)	$\frac{1}{4} \times 2\pi = \frac{1}{2}\pi$ $= 0.5\pi$ $= 1.57 \text{ rad [B1: answers in terms of } \pi \text{ also accepted]}$
	(b)	Since weight and tension provide for centripetal force [B1] $T + W = m \frac{v^2}{R}$ $v = \sqrt{\frac{R}{m}(mg + T)}$ $= \sqrt{\frac{1.5}{0.5}((0.5)(9.81) + 10)} \quad [\text{B1}]$ $= 6.6869$ $\approx 6.7 \text{ m s}^{-1}$
	(c)	By the principle of conservation of energy, loss in GPE = gain in KE $mgR = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$ $v_B = \sqrt{2gR + v_A^2}$ $= \sqrt{2(9.81)(1.5) + (6.7)^2} \quad [\text{C1}]$ $= 8.6107 \text{ m s}^{-1}$ $\approx 8.6 \text{ m s}^{-1} \quad [\text{A1}]$
	(d)	$v_B = R\omega_B$ $\omega_B = \frac{v_B}{R}$ $= \frac{8.6107}{1.5} \quad [\text{C1}]$ $= 5.7404$ $\approx 5.7 \text{ rad s}^{-1} \quad [\text{A1}]$
	(e)	Weight acting vertically downwards and tension acting leftward. [B1]
	(i)	Note: no need to compare lengths since force magnitudes have not been calculated
	(ii)	Since the <u>net force has a downward component while the bob travels upwards</u> OR Since the <u>acceleration has a downward component while the bob travels upwards</u> OR Since <u>kinetic energy is being converted to gravitational potential energy</u> OR Since <u>weight is doing negative work on the bob</u>

4

			<u>OR</u> Since <u>weight acts opposite to the velocity [B1]</u>
--	--	--	---

4	(a)	The vibration of the lattice ions increased [B1] thus increasing the collisions
---	-----	---